

E-commerce Sales Data Analysis

1. Data Entry & Organization

Data was segmented into four distinct tables: CUSTOMER, PRODUCT, STORE, and SALES.

2. Data Cleaning & Transformation

- Corrected formatting errors in text fields wherever necessary.
- Standardized date formats for the Order_Date column.
- Handled missing values where necessary.
- Removed unnecessary spaces and corrected inconsistent column names.
- Standardized the data by converting text to uppercase, formatting percentages, and converting decimal values to whole numbers.

These transformations improved data quality and prepared the dataset for analysis.

3. Excel Formulas & Functions

SUM Functions

Task	Formula Used
Total Quantity of products sold	SUM
Unit price for COD orders	SUMIFS
Total revenue generated from Credit Card payments	SUMIFS
Total sales generated by PayPal transactions	SUMIF
Total discount amount given across all orders	SUM

IF Function

Task	Formula Used
Created column as High/Low for Total Amount greater than 1500	IF
Created column as Order Value for below	IFS
• Premium – Greater than 2000	

- High – Greater than 1000
- Normal – Less than 1000

Created column as Price for Discount >15% and Total Amount >1000 IF and AND

Average Functions

Task	Formula Used
Average of Unit Price	AVERAGE
Average Quantity Sold per Order	AVERAGE
Average of Total Amount for orders where Payment Type is PayPal	AVERAGEIF
Average of Total Amount for orders where Payment Type is Credit Card and Quantity > 2	AVERAGEIFS

Text Functions

Task	Formula Used
Merge Sale_ID and Quantity	CONCAT
Extract Year from Order_Date	YEAR
Convert Month to Text	TEXT

VLOOKUP – Implemented to fetch customer details (Customer Name, Loyalty_Level) and product details (Product_Name, Category, Cost) from related tables.

4. Descriptive Statistics

A statistical overview was conducted to understand the scale of the business operations.

Calculated Sales Performance in the Sales Table and Inventory Valuation in the Product Table using Descriptive Statistics.

5. Data Summarization (Pivot Tables)

- Category Wise Total Sales – Identified which product categories drive the highest revenue.
- Distribution of Orders – Analyzed the distribution of orders based on quantity purchased per order.
- Yearly Total Sales – Highlighted the most commonly used payment types by customers year wise.

- **Monthly Total Sales** – Analyzed how sales performance fluctuates across different months.
- **Store Comparison** – Compared monthly sales performance across different store locations.
- **Sparklines** – Integrated mini-trendlines to show individual store performance at a glance.

6. Data Visualization

- **Pie Chart** – Visualizes Category-wise Total Sales showing Sports (25%) and Electronics (23%) as leading categories.
- **Column Chart** – Illustrates Year-wise Total Sales by Payment Type.
- **Line Chart** – Shows the Monthly Total Sales Trend.
- **Scatter Chart** – Used to analyze distribution of orders based on quantity.
- **Interactive Slicers** – City-wise slicer allows filtering the Pie, Column and Line chart dashboard by city.

Key Business Insights and Findings

1. Category-wise Sales

- Sports and Electronics categories together contribute nearly 48% of total sales, making them the primary revenue drivers for the e-commerce platform.
- Beauty and Clothing categories generate comparatively lower sales share, indicating potential opportunities for targeted promotions or product diversification to improve category performance.

2. Monthly Sales Trend

- Sales show seasonal fluctuations, with noticeable declines during certain months and recovery toward the later part of the year, suggesting demand variability across the calendar.
- The sharp drop observed in one of the months indicates a possible seasonal or operational gap, highlighting the need for promotional campaigns during low-performing periods.

3. Payment Type vs Total Sales

- Credit Card transactions consistently generate the highest sales contribution, indicating strong customer preference for digital payment methods.
- COD and PayPal transactions remain relatively stable across the years, suggesting a balanced distribution of payment options without heavy dependence on a single payment channel.

4. Quantity-wise Orders

- The majority of orders are concentrated around low purchase quantities (1–2 items), indicating typical consumer purchasing behavior focused on smaller baskets.
- Order frequency declines as quantity increases, suggesting potential opportunities for bundle offers or quantity-based discounts to increase the average order value.

Technical Skills Demonstrated

Tools: Microsoft Excel, Power Query

Functions: VLOOKUP, SUMIFS, COUNTIFS, IF/IFS, TEXT, YEAR

Techniques: Pivot Tables, Charts, Data Visualization

Final Dashboard

All charts are formatted with professional colors and a clean layout, featuring a city-wise Slicer that allows interactive analysis by automatically updating the Category, Trend, Payment charts.

